

Bitter Type DC Electromagnet System

High-Field Air-Core Electromagnet with Advanced Thermal Management



Precision Magnetic Field Control.
Stable Bipolar Power Delivery.
Built for Research Applications.

The CRYONANO Bitter Electromagnet is a high-field, air-core DC magnet engineered for applications requiring intense magnetic fields and superior thermal stability. Based on the classical Francis Bitter design, the system uses stacked copper plates with precision cooling channels to handle extremely high currents while maintaining structural and thermal integrity.

Designed for demanding research and industrial environments, this system delivers reliable, scalable, and customizable magnetic field solutions for advanced physics and high-power applications.

Notes

- Heat generation scales with square of magnetic field, requiring aggressive cooling
- High-current operation demands robust electrical and thermal infrastructure
- Designed for continuous operation under controlled conditions
- Fully engineered-to-order system based on application

KEY FEATURES

- **High Magnetic Field Generation**
Generates magnetic fields up to 11 Tesla for high-performance research applications.
- **Air-Core Design**
Eliminates ferromagnetic saturation limits for stable high-field operation.
- **Bitter Plate Construction**
Optimized plate architecture ensures superior heat dissipation during operation.
- **High Current Operation Support**
Designed for very high current operation (~1000 A and above).
- **De-Ionized Water Cooling**
High-flow de-ionized water cooling enables efficient thermal management.
- **Symmetric Field Profile**
Provides highly stable and uniform magnetic field distribution.

APPLICATIONS

- Gyrotron systems
- Accelerator physics
- Plasma research
- High-power microwave systems
- Advanced material research
- Magnetic field testing and characterization

EMC2T Electromagnet System

SPECIFICATIONS

MAGNET SPECIFICATION

Magnet Type	Air-Core DC Bitter Electromagnet
Construction	Stacked copper Bitter plates with insulating spacers
Bore Diameter	65 mm
Axial Length	300 mm
Maximum Central Field	Up to 1.1 Tesla
Field Profile	Peak at center, symmetric axial decay
Coil Configuration	Dual concentric coils, series connected
Operating Current	~1000 A (application dependent)
Continuous Operation	Yes

COOLING SPECIFICATION

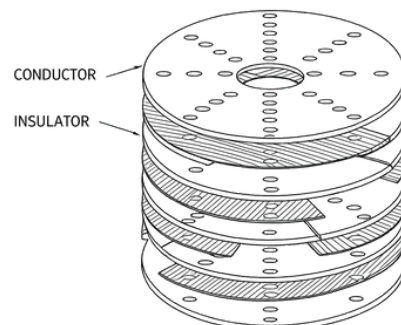
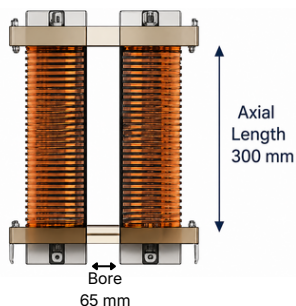
Cooling Method	De-ionized water cooling
Cooling Flow Direction	Axial through Bitter plate channels
Cooling Water Temperature	~10°C
Cooling Water Pressure	Up to 12 bar
Cooling Design	High-velocity flow through aligned channels

POWER SUPPLY SPECIFICATION

Input Supply	415 V, 50 Hz, 3-phase, 4-wire AC
Output Type	Regulated DC
Current Capability	~1000 A and above
Nominal Power	30 – 40 kW
Maximum Power	Up to 50 kW
Mode	Constant Current (optimized for magnet operation)

CONSTRUCTION AND MATERIALS

Bitter Plates	High-conductivity copper
Insulation	High-strength insulating spacers
Cooling Channels	Precision-drilled aligned holes



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